Learning Tonal Processes of Different Computational Complexity

Presenter: Zijing Xu

Rutgers University

Workshop on Subregular Morphophonology 2026

Typology and learnability

- Formal complexity and phonetic substance biases could cause simple or phonetically-motivated patterns to accumulate, thus producing a corresponding skew in natural-language typology (Moreton & Pater, 2012; Wilson, 2003, 2006; Finley & Badecker, 2007; Becker, Ketrez, & Nevins, 2011).

- A learner with formal complexity bias would acquire simpler patterns faster or better than complex ones.

This study hypothesizes that some logically possible phonological patterns are unattested in typology because of the restrictions on human's phonological learners. In other words, they are beyond the complexity that human can learn in phonology.

Typology and learnability

Why phonological processes?

- Most previous experimental work on phonological learnability focuses on phonotactic patterns with different computational complexities (e.g., Lai, 2013, 2015; McMullin & Hansson, 2019).
- The phonotactic patterns and phonological processes are different.

Why tones?

- Most previous long-distance learnability studies focus on consonantal and vowel harmony (e.g., Lai, 2012, 2013; Finley, 2012, 2015; McMullin & Hansson, 2019).
- Segments and tones are different (Yip, 2002; Hyman, 2011; Jardine, 2016).

This study investigates

- What kinds of computational complexity of tonal processes are learnable by human beings?
- Does the existing framework capture the learnability differences?

Subregular hierarchy of functions

Phonological processes can be modeled with functions in Formal Language Theory (Heinz, 2018; Jardine, 2025).

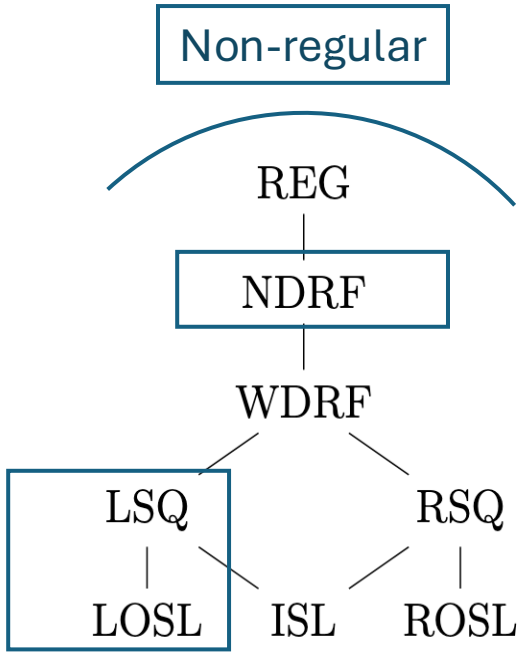
A function f is a map from members of some set X (domain) to members of a set function domain Y (co-domain) such that each member of X is mapped to exactly one member of Y .

When defining a function, we can specify its domain and co-domain by writing:

$$f: \Sigma^* \rightarrow \Gamma^*$$

Σ* denotes all strings that can be constructed from symbols in Σ

Γ* denotes all strings that can be constructed from symbols in Γ



Subregular hierarchy of functions (Heinz, 2018)

Majority Rules (MR) Harmony

Description

If the number of segments with (feature F) [αF] is greater than the number of segments with [−αF], then segments with [αF] are the triggers of harmony and segments with [−αF] are the targets and undergo change (Lombardi, 1999; Baković, 2000; Heinz & Lai, 2013).

Why is it 'pathological'?

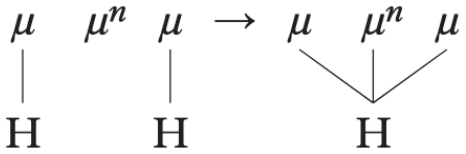
- MR is logically possible, which can be generated in classic OT with AGREE (F) ≫ IDENT (F).
- However, it is unattested as we know and considered phonologically bizarre.

Complexity of MR

Non-regular (Riggle, 2004; Heinz and Lai, 2013)

Unbounded Tonal Plateauing (UTP)

Description

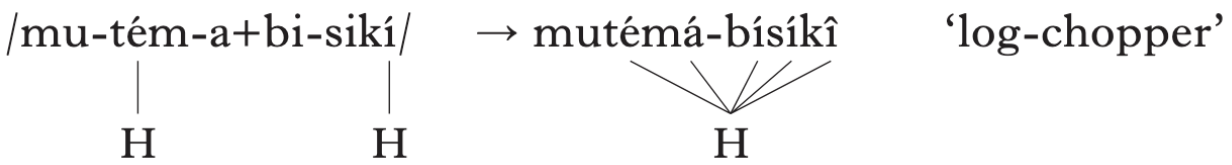


Any number of tone-bearing units between two underlying high tones become high (Jardine, 2016).

Typology

Unbounded circumambient process has been attested both in tonal (Hyman, 2011; Jardine, 2016) and segmental domains (McCollum et al., 2020).

Case: Luganda



'log-chopper'

Complexity of UTP

Non-deterministic (Jardine, 2016; McCollum et al., 2020)

High Tone Spreading (HTS) (Left-to-Right)

In typology, tone spreading tends to be perseverative rather than anticipatory, and the most likely tone to spread is /H/ (Hyman, 2017).

A tonal learnability study shows that HTS left-to-right is learned better than right-to-left across languages (Kwong, 2026).

Typology

Case: Copperbelt Bemba (Bickmore & Kula, 2013)

(17)	a. bá-ká-fík-á	'they will arrive'	/bá-ka-fik-a/
	b. bá-ká-mú-lóóndólól-á	'they will introduce him/her'	/bá-ka-mu-londolol-a/
	c. bá-lá-mú-lúk-íl-á	'they weave for him/her'	/bá-la-mu-luk-il-a/
	d. tù-kà-páápáátík-á	'we will flatten'	/tu-ka-páapaatik-a/
	e. tù-là-páápáátík-á	'we flatten'	/tu-la-páapaatik-a/
	f. ú-kú-lóóndólól-á	'to introduce'	/ú-ku-londolol-a/
	g. ú-mú-límí	'farmer'	/ú-mu-limi/

Complexity of HTS (L-to-R)

Left-subsequential

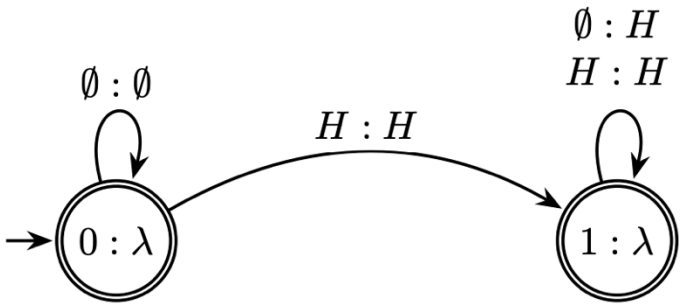
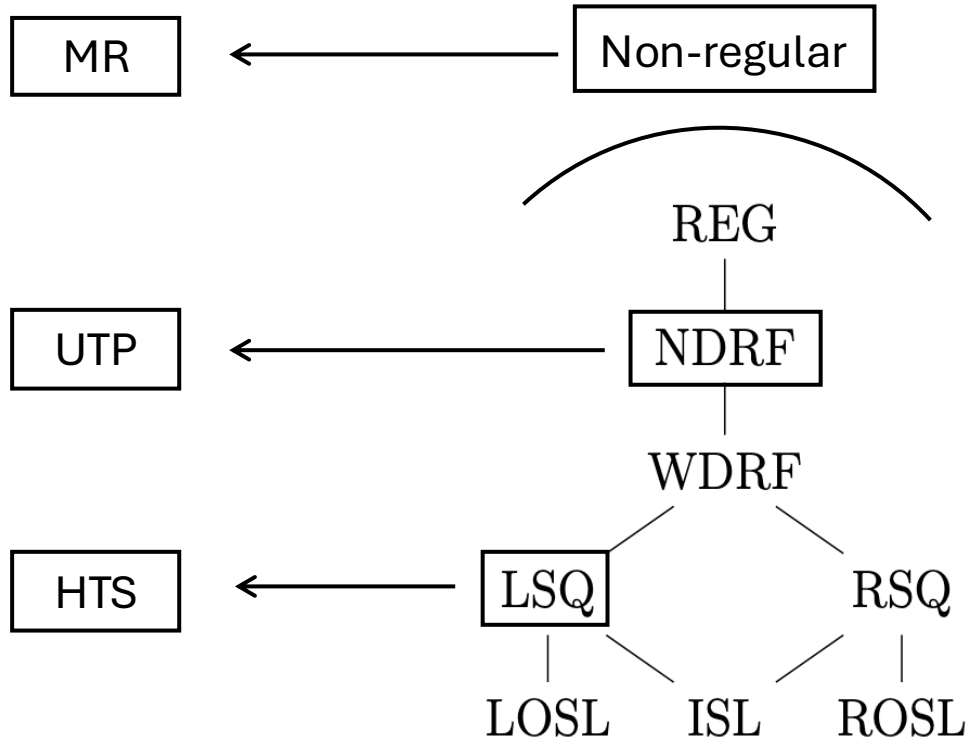


Figure 1. A Left subsequential for unbounded HTS in Bemba. Underlying unspecified and surface L tones are unmarked (∅).

Three processes in subregular hierarchy of functions

	UR	MR	UTP	HTS
a.	/+ - -/	[- - -]	[+ - -]	[+ + +]
b.	/- + +/	[+ + +]	[- + +]	[- + +]
c.	/- - -/	[- - -]	[- - -]	[- - -]
d.	/- + -/	[- - -]	[- + -]	[- + +]
e.	/+ - +/	[+ + +]	[+ + +]	[+ + +]
f.	/+ + +/	[+ + +]	[+ + +]	[+ + +]

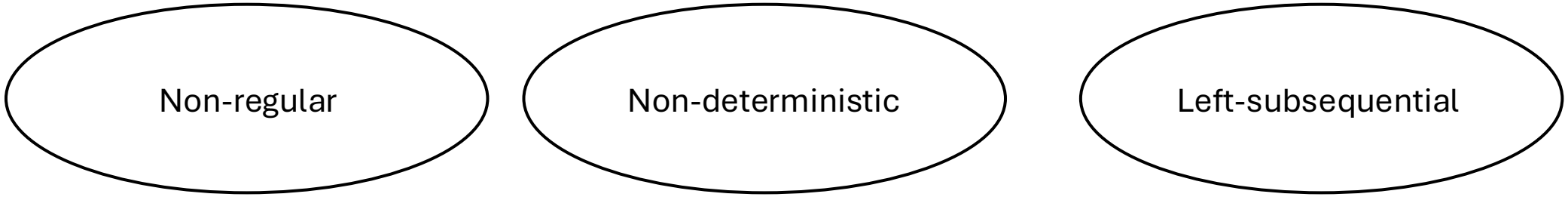
Table 1. Example mappings of underlying representation (UR) given by MR, UTP and HTS



Subregular hierarchy of functions (Heinz, 2018)

The three processes differ crucially in **computational complexity** and **typology.**

Prediction of learnability



Process	Majority rules	Unbounded tonal plateuing	High tone spreading (L-to-R)
Typology	✕	✓	✓
Complexity	★★★	★★	★
Learnability	✕	✓ (worse)	✓ ✓ (better)

Methodology

The artificial language learning paradigm (e.g. Folia et al, 2008; Gómez & Gerken, 2000; Petersson et al., 2004; Reber, 1967)

Stimulus

- Cantonese tonal inventory
  - H level tone (T1), L level tone (T6)
- Cantonese syllable inventory
  - 1. real syllable which can bear T1 and T6 (33 syllables)
  - 2. five-syllable pseudo words
- Acoustic control using Praat
  - Token duration normalization: 650ms
  - silent interval: 5ms
  - Mean F0: H level tone: 240 Hz, L level tone: 190 Hz

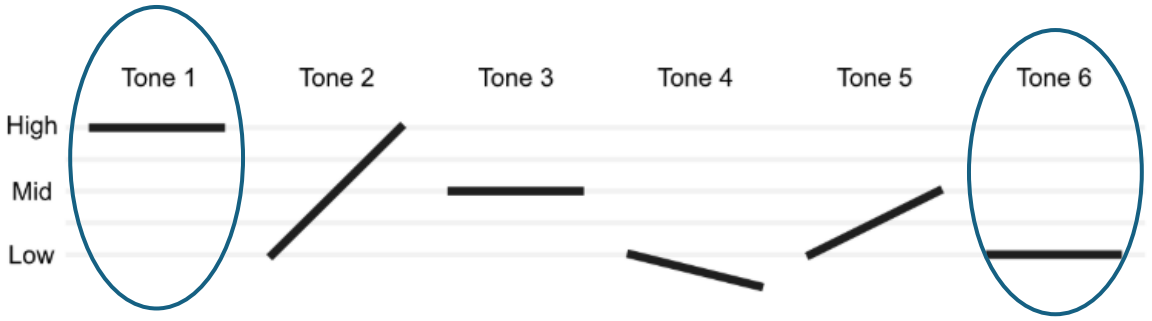
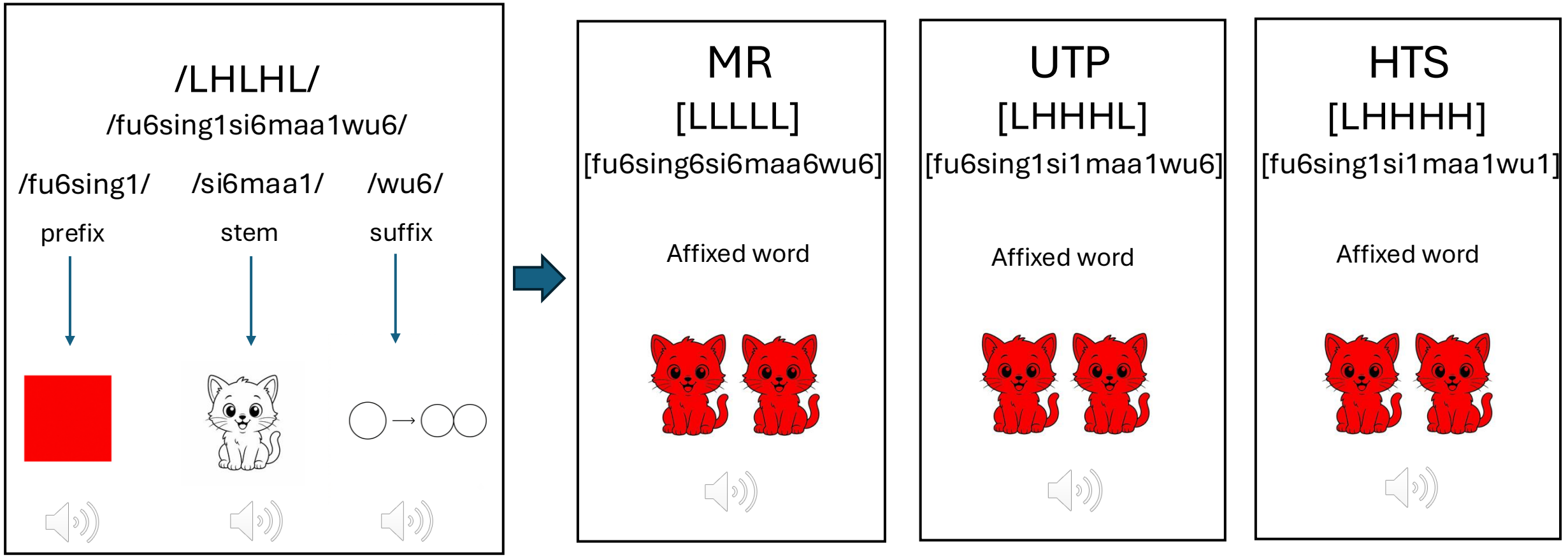


Figure 2. Cantonese lexical tones

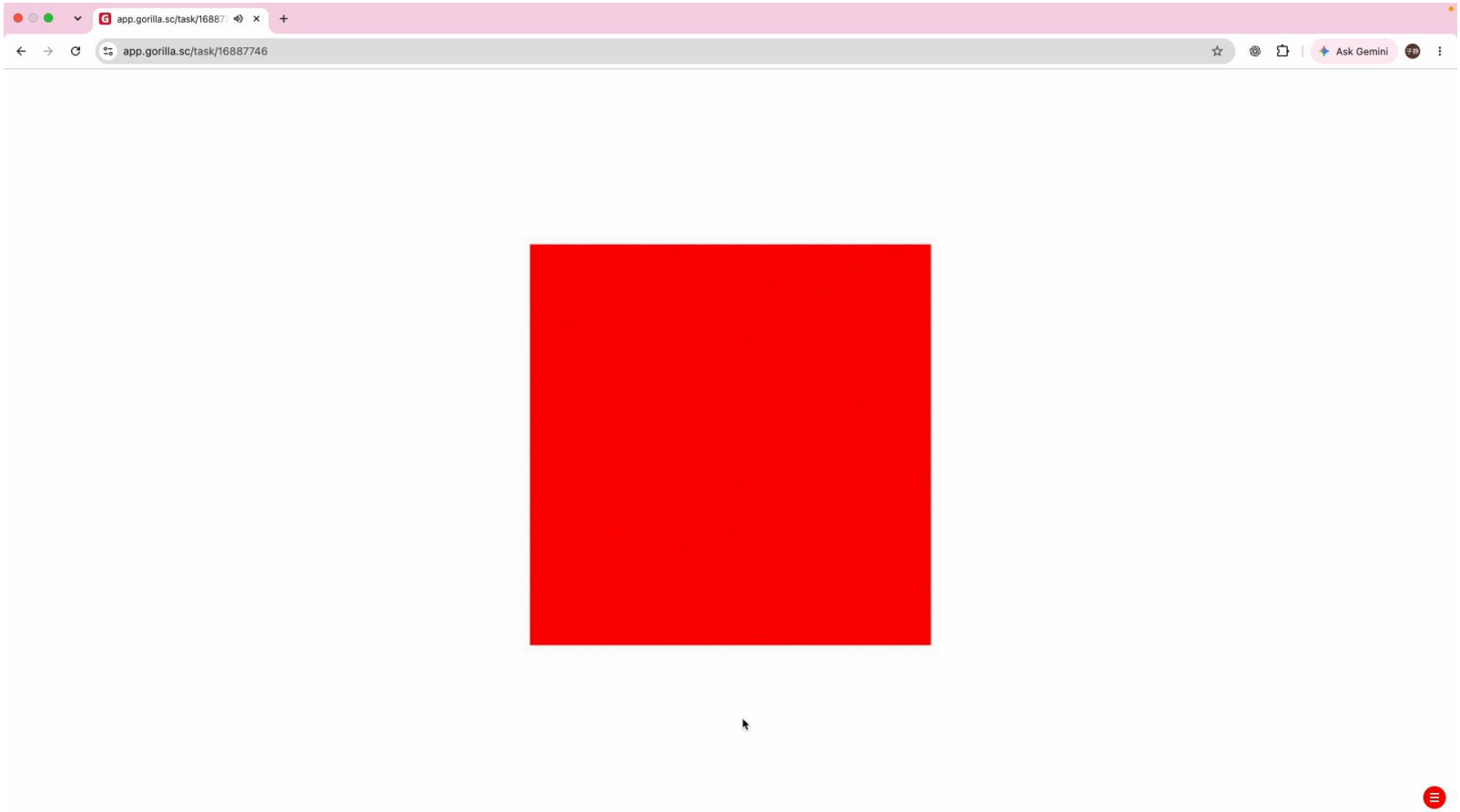
Methodology

- Word learning paradigm Example



Methodology

- Tasks: grammaticality judgment and report how certain they are using a continuous confidence slider Example



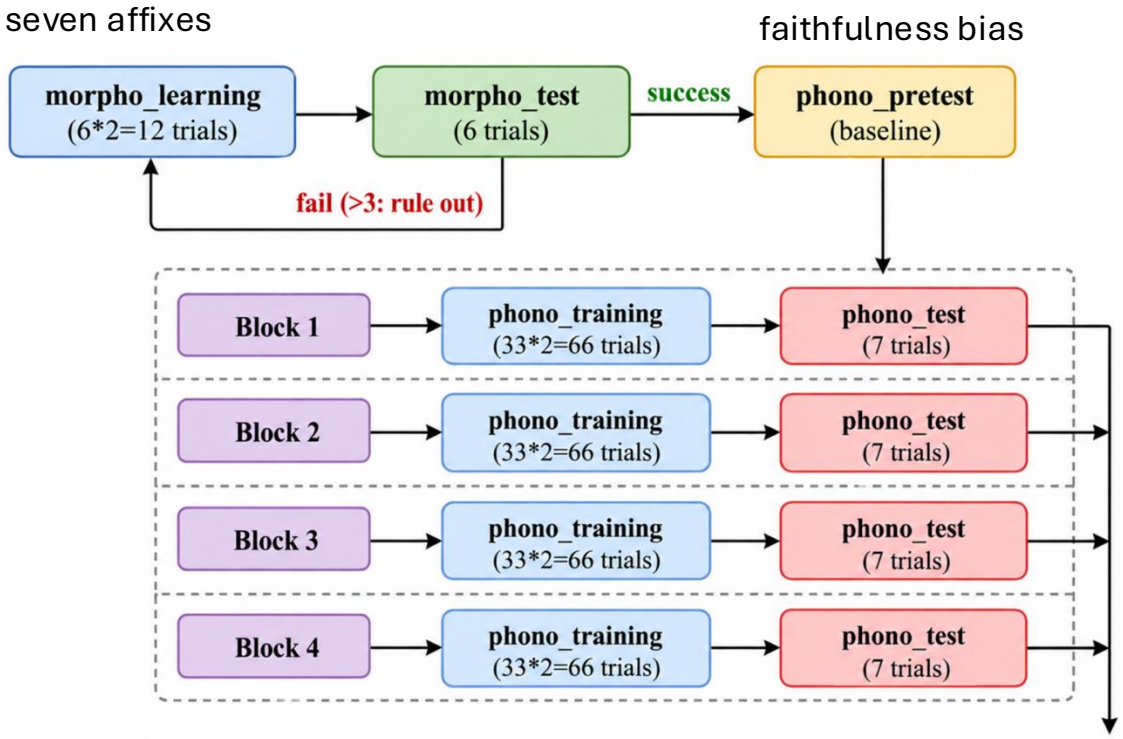
Methodology

- Participants
  - So far, 12 native speakers of Cantonese adults in Hong Kong were recruited online in each condition. N=36 in total.

- Procedure
  - The phonological training phase

    264 words in total

    approximately 35 minutes
  - The phonological test phase
    - stems are different from training
    - All three conditions used the same 28 trials
    - In each condition, half grammatical, half ungrammatical



Preliminary results

A binomial generalized linear mixed-effects model was fitted.
Accuracy ~ Condition × Session + (1 | Subject) + (1 | Item)

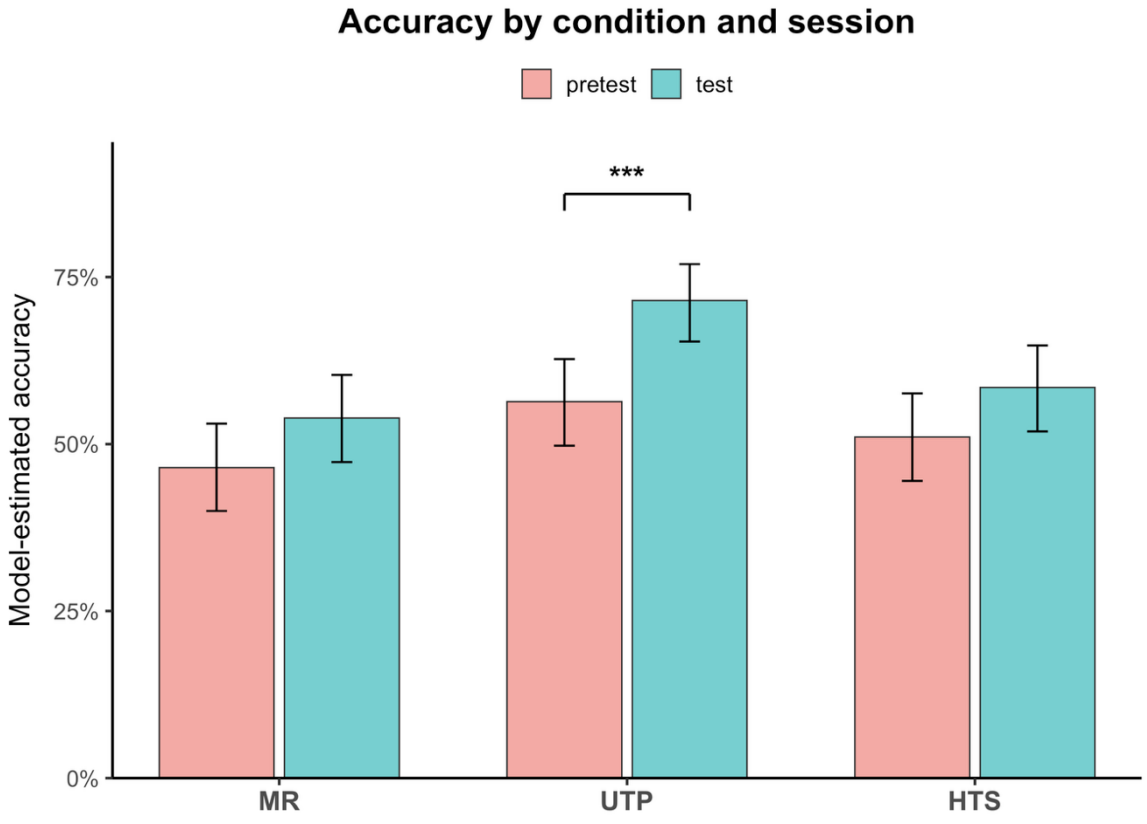


Figure3

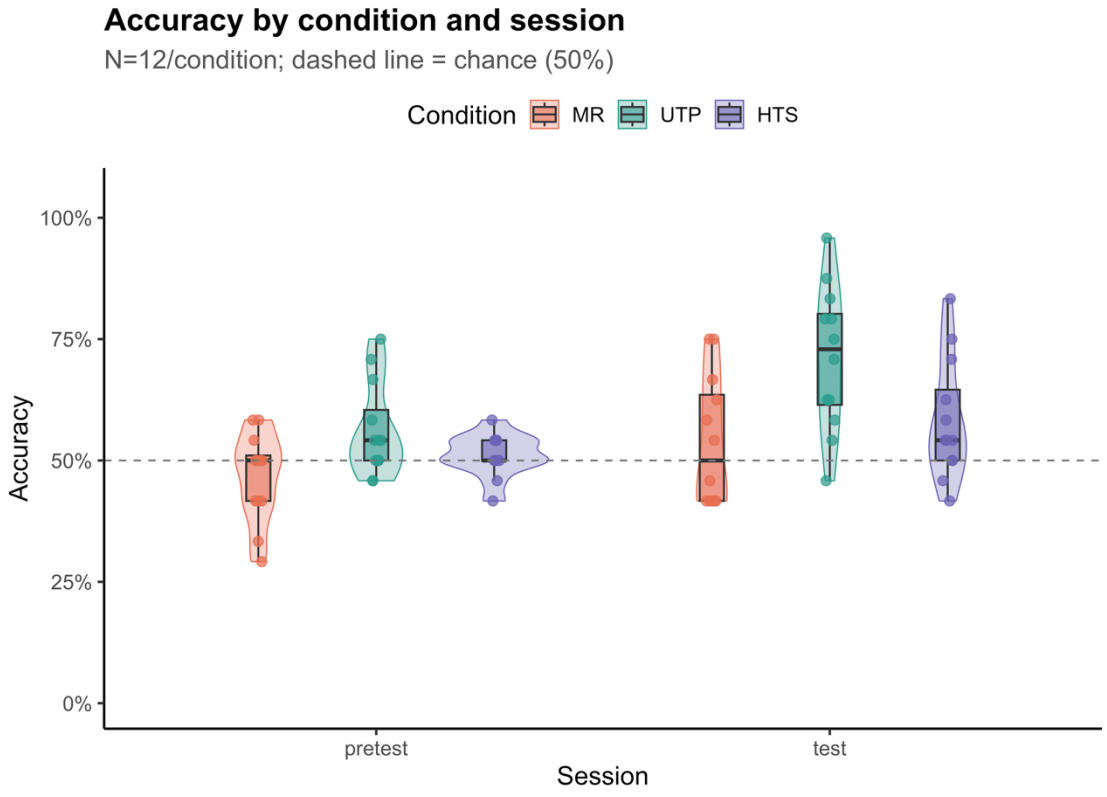


Figure 4

Condition × Session interaction was not statistically significant so far: UTP vs MR, β = 0.366, *p* = .135,
HTS vs MR, β = 0.002, *p* = .992.

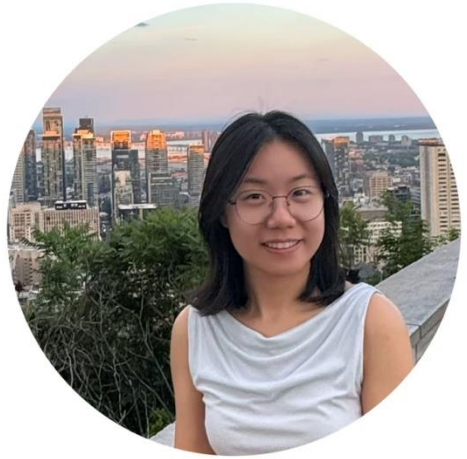
“Early” conclusion

- So far, MR is learned the worst, HTS is learned intermediate, whereas UTP is learned the best across conditions.
- Ongoing research:
  1. 30 participants for each condition
  2. check other factors which may influence the learnability results and individual differences, e.g., confidence scores, working-memory capacity.

Acknowledgment



Prof. Regine Lai



Dr. Chunzi Li



Mijia Ou



Dr. Ken Kwong

香港中文大學
The Chinese University of Hong Kong

This research is supported by the Departmental Fund for Research, Department of Linguistics and Modern Languages, The Chinese University of Hong Kong, awarded to Regine Lai.

Thank you!
Any question and suggestion are welcome.

If you have more suggestions, please contact zijing.xu0323@rutgers.edu.